

# MOVIE RECOMMENDATION SYSTEM

GROUP 3 - KARU

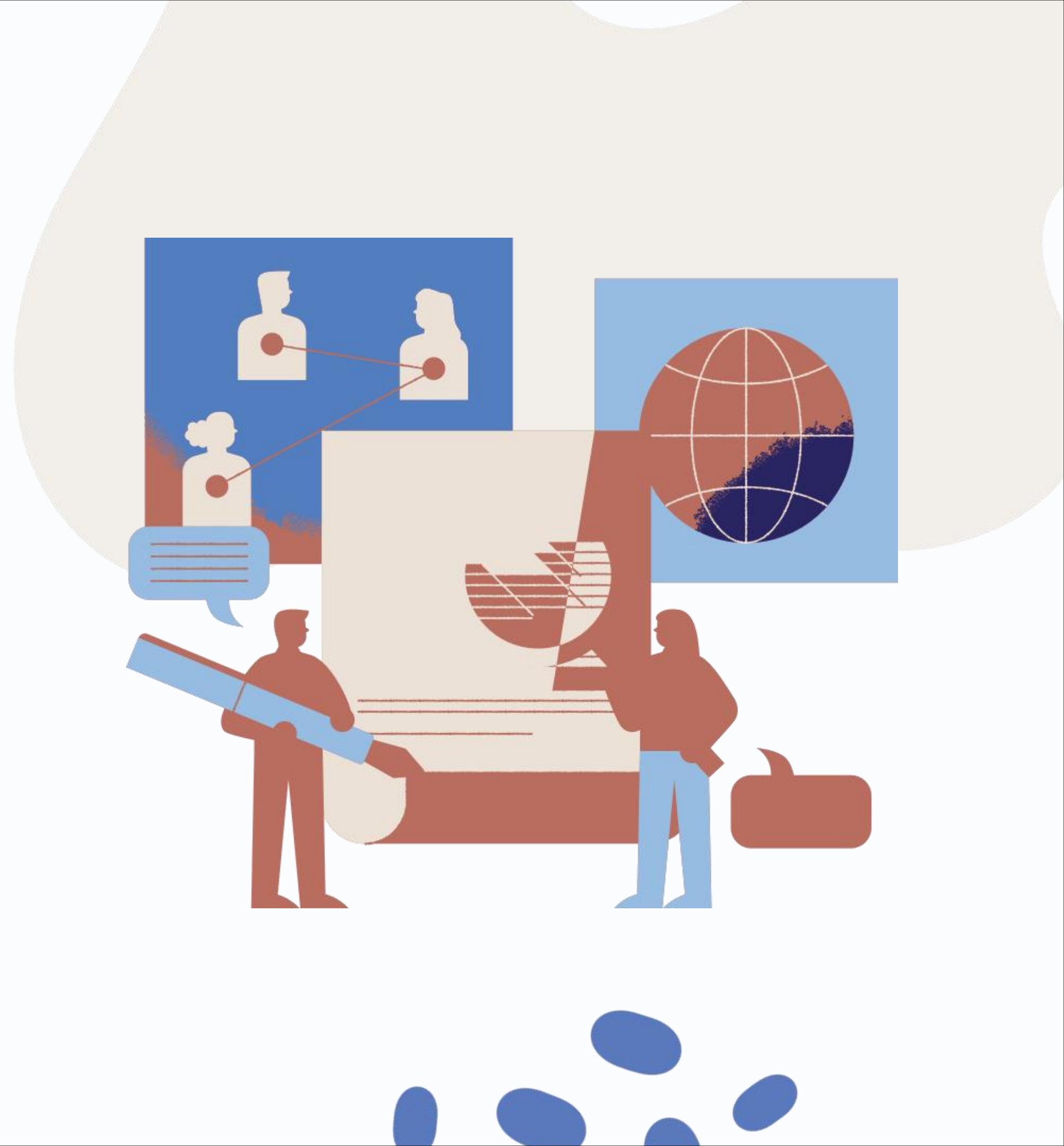
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# BUSINESS UNDERSTANDING

## Stakeholders

Our customer is LumeStream, a tech company dedicated to making movie watching more personal and fun. Its streaming service is also called LumeStream.

## Problem statement

LumeStream service users are exposed to a vast number of movies making it difficult for them to identify movies that match their interests and preferences.

## Objectives

Build a model that provides top 5 movie recommendations to a user, based on their ratings of other movies.

# DATA UNDERSTANDING

## Steps taken:



01

### Data Loading

In the Data Loading step, we loaded the datasets into pandas to create data frames that we used in the subsequent processes of the project.

02

### Data Inspection and Cleaning

We roughly inspect the our data and look for errors such as missing values and duplicates. We handle the errors by filling in missing values with averages or most frequent value and dropping duplicates.

03

### Data Visualization and Analysis

With the aid of charts and other tools, we analysed data and came up with insights and recommendations.

# DATA PREPARATION

Before modeling, the data was cleaned and structured so the recommendation system could accurately learn user preferences.





# MODELLING

Steps taken during modelling:



01

## **Creating a simple model**

Started with a simple model

02

## **Creating a different recommendation model**

Created another type of recommendation model that is generally more suitable for the project objective

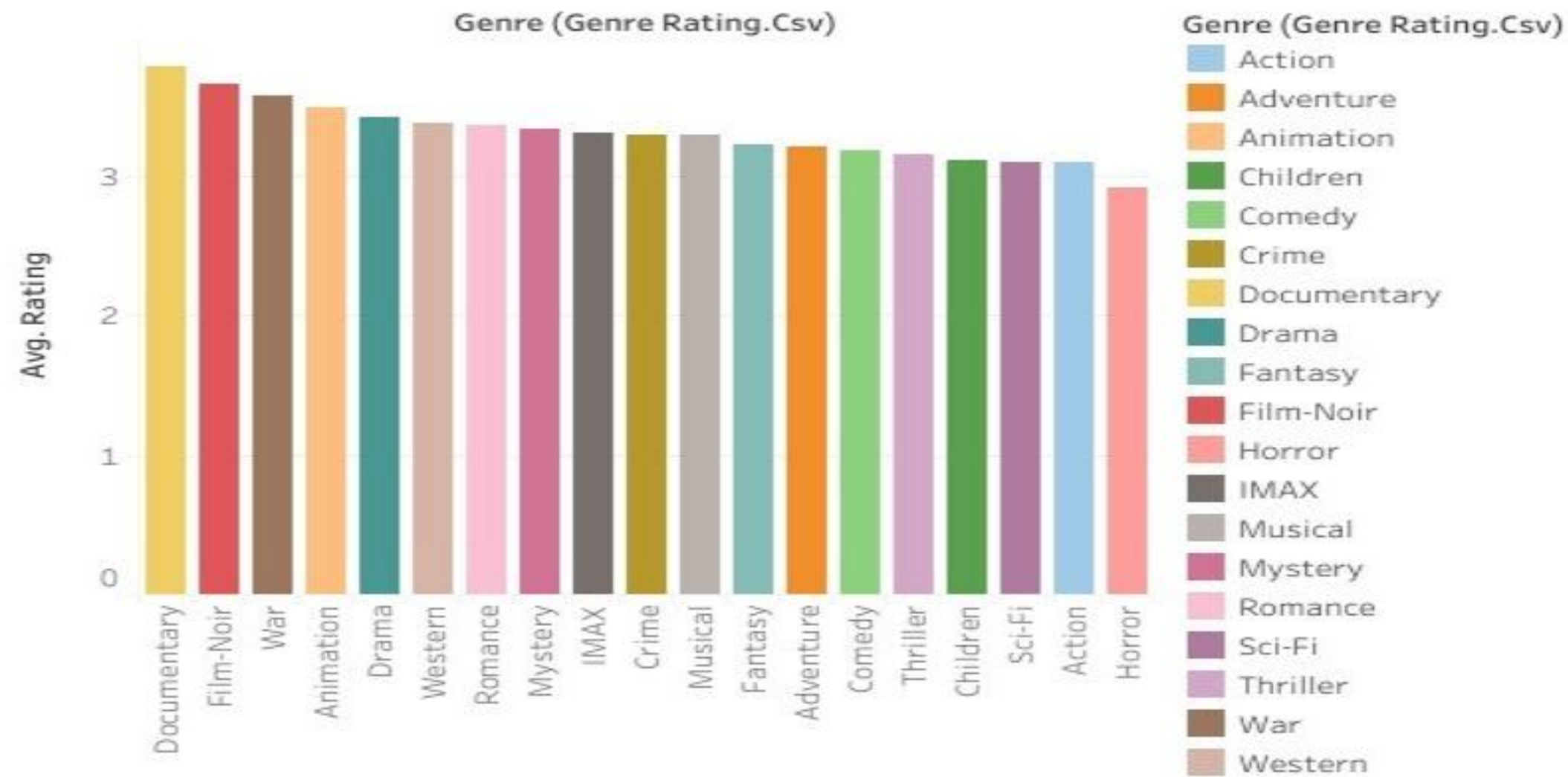
03

## **Improving the model**

We compare the two and improve it so it can meet our objectives.

# VISUALIZATIONS

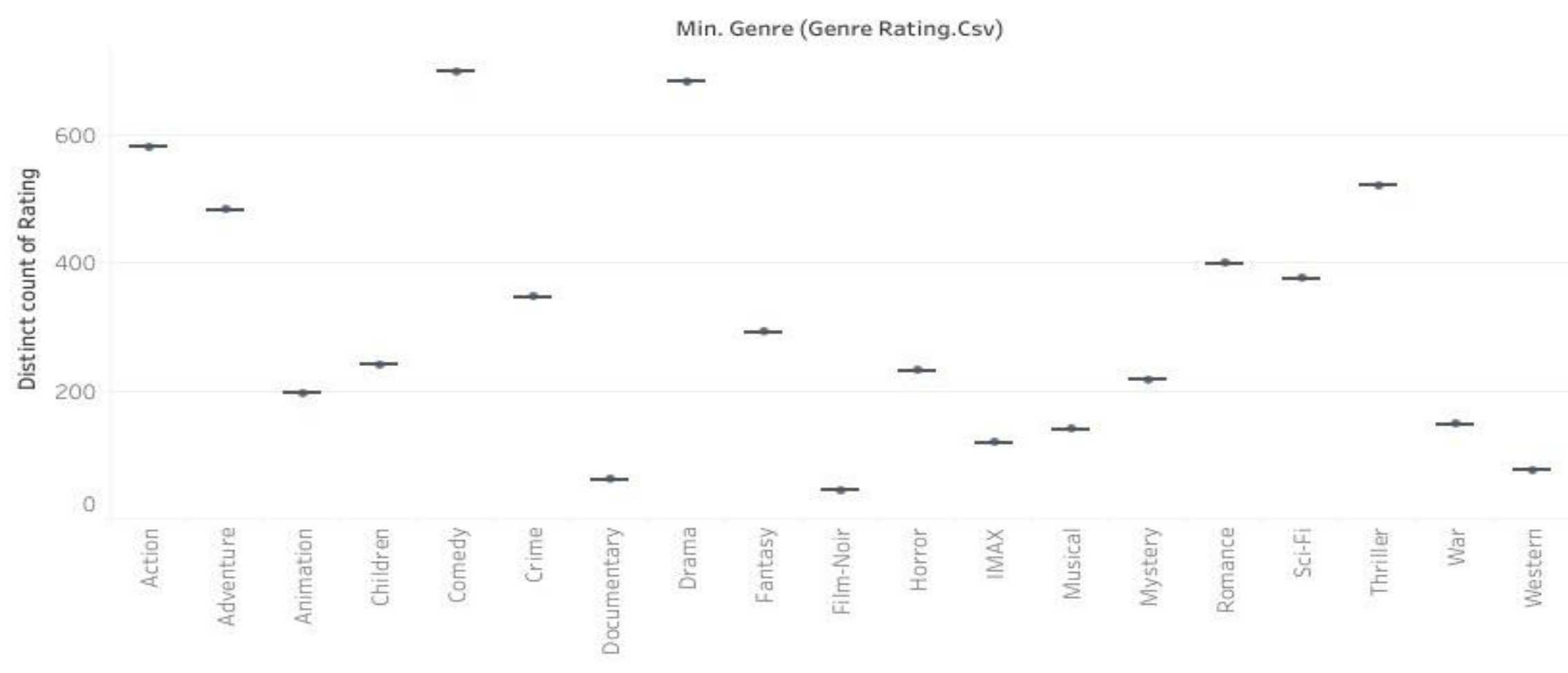
## AVERAGE RATING PER GENRE



- \* We can see the genre ratings do not vary by much.
- \* We can see documentary is the highest rated genre with 3.8
- \* We can see horror is the lowest rated genre with 2.9

# VISUALIZATIONS

BOX AND WHISKER PLOT OF THE GENRES

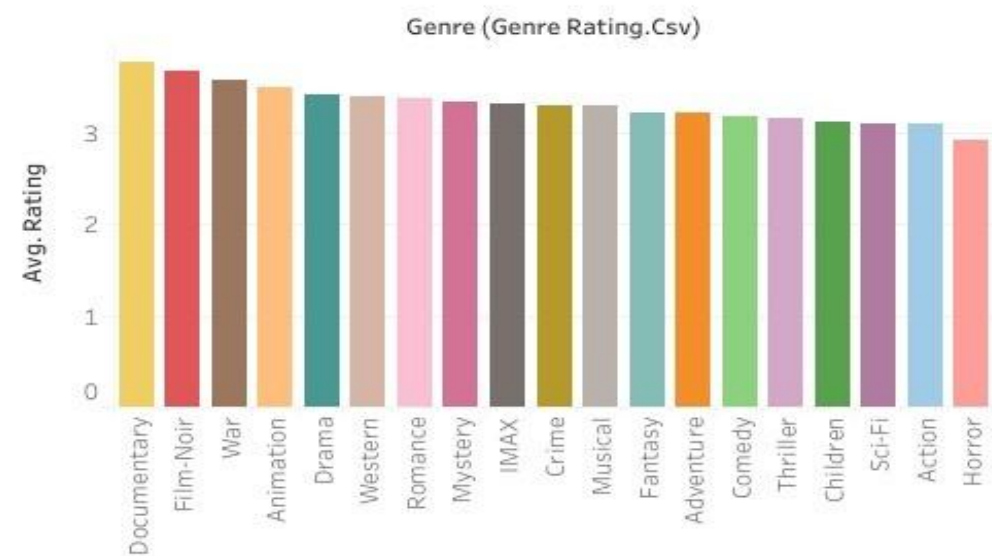


Comedy and Drama have the highest number of distinct ratings, making them the most represented genres in the dataset.  
Film-Noir, Documentary, and Western have the lowest counts, indicating limited representation.  
However, the chart does not show rating variability because each genre is represented by a single aggregated value.

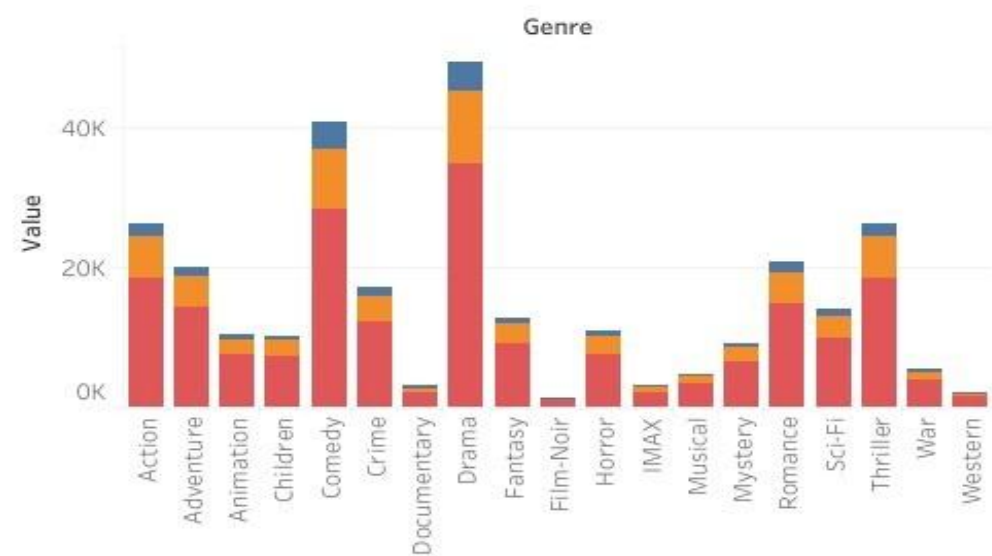
# VISUALIZATIONS

## PROJECT DASHBOARD

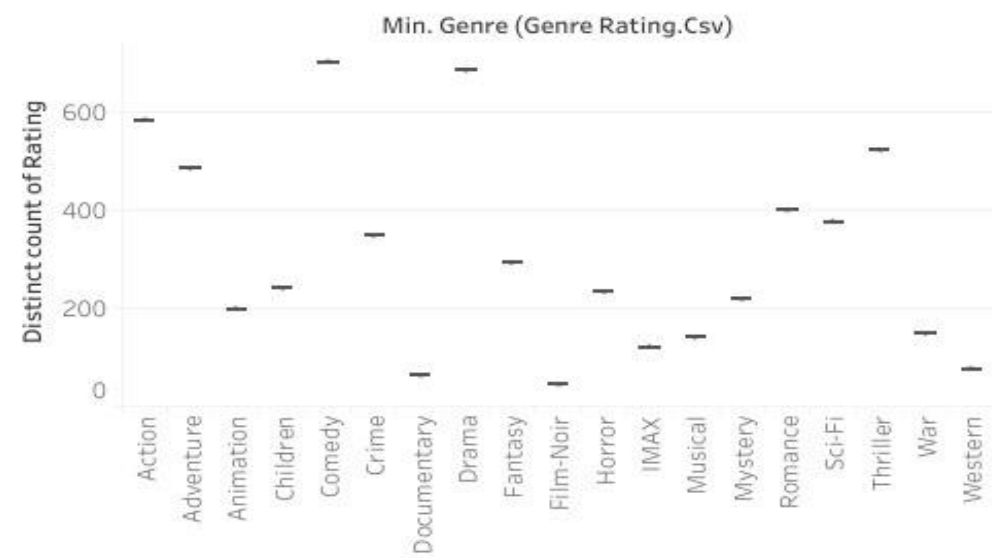
### AVERAGE RATING PER GENRE



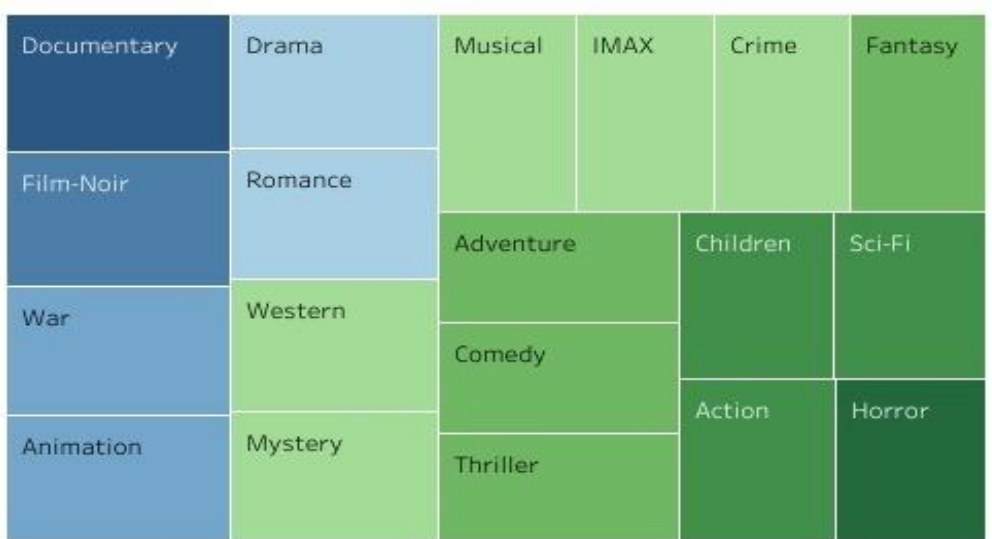
### MEASURE VALUES PER GENRE



### BOX AND WHISKER PLOT OF THE GENRES



### TREE MAP SHOWING THE GENRES





# RECOMMENDATIONS



01

The model is now ready to be integrated to the system.

02

After collecting more user data, the company could build more complex but better performing models

# CONCLUSION

**The recommendation system aims at improving customers experience by providing personalized movie suggestions.**

**The system was able to recommend relevant movies effectively.**





*THANK  
YOU!*